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EEG-CAR-01 Rev C – work report

Not a controlled document. It is the report on the work of 8 September 2026 that ECO-EEG-030 to ECO-EEG-033 record, and it says what was run, what the repository contradicted, what was assumed, and what is still open. **Licence:** CC BY-SA 4.0

1. What was run, and what it printed

1.1 Baseline, on the untouched tree

Command	Result
<code>tools/emit_all.py --cached</code>	run. 0 violations, 145 of 145 nets connected, 3 745 segments, 552 vias, minimum clearances 0.260 / 0.285 / 0.285 / 0.275 mm. Every text output reproduced byte for byte against the committed files; only the PNG, PDF and zip outputs differed, on embedded timestamps
<code>tools/simulate_production.py</code>	run. 186 passed, 0 failed, 5 open – not the 193 / 0 / 6 the register stated. See 2.1
<code>kicad-cli --version</code>	NOT RUN. KiCad is not installed on this machine and could not be installed. No ERC and no DRC result is claimed anywhere in this work

Four Python packages had to be installed before anything would run at all: shapely, scikit-image, numpy (present) and trimesh. **cadquery could not be installed:** the machine had 472 MB of free disk and cadquery-ocp needs several times that. `tools/mech_gen.py` imports cadquery at module scope, so it cannot even be imported here, and **no printed part was regenerated in this work.**

1.2 The Rev B geometry under the new rule set

`tools/grade_revb.py`, on the released `kicad/EEG-CAR-01_RevB_routed.kicad_pcb` and `tools/routed.pkl`, census checked first (3 745 segments, 552 vias):

Finding	Rule	Count
1	vias inside an SMD pad	67
2a	dangling track ends	5
2b	redundant copper loops outside a plane	221 , on 22 nets
3	angle below 90 degrees between two segments	171

Finding	Rule	Count
3	segments off a 0/45/90/135 degree axis	158
4	off-centre pad entries	299
5	conductors under their class minimum	322
5	net classes on a forbidden layer	50
5	vias on a class that forbids them	18
	total	1 311

The Rev B DRC report is **not withdrawn**. It measured the twelve rules it had, correctly.

Two counts in `tools/drc_geometry.py` were wrong before they were right, and both are recorded in the source rather than quietly fixed. The loop check first modelled copper as touching polygons and reported **3 356** loops, where three mutually touching polygons are not a loop; re-modelled topologically – junctions as nodes, segments as edges, T-junctions split – it reported **476**; excluding degenerate edges, where a segment's two ends land on the same via, it reports **221**. The first two numbers are wrong and are not quoted anywhere else.

1.3 The schematic against the design source

`tools/sch_netlist.py` parses the emitted `.kicad_sch` files back – pin geometry from their own embedded `lib_symbols`, wires, junctions, labels, hierarchical labels, power symbols – and diffs. It does not ask the generator what it meant.

	<code>design.py</code>	schematic
nets	156	156
pins	614	614
structural problems	–	0
netlist differences	–	0

It found four real errors on its first run: R50, R51, R52 and R94 were wired by geometry rather than by pin number and had pin 2 where `design.py` has pin 1, and R94/R95 were drawn as a divider instead of two I2C pull-ups. A resistor is symmetric, so the circuit was right and the netlist was wrong. Both are fixed and the helper that caused it now wires by pin number.

1.4 ERC, DRC and the collision check

Check	State
<code>kicad-cli sch erc</code>	NOT RUN. KiCad is not installed
<code>kicad-cli pcb drc</code>	NOT RUN. Same

Check	State
structural ERC (tools/sch_netlist.py)	run: 0 findings. Every pin on a net; no designator placed twice and every unit of a multi-unit part placed once; no accidental single-pin net; every power net driven by a power symbol
board read-back (tools/emit_revc.py)	run: passes. 211 footprints, 636 pads, 156 nets, 0 segments, 0 vias , 37 locked, netlist identical to design.py, every rule area present under the name the .kicad_dru uses, project classes and all 156 assignments complete, track_dangling at error
footprint audit (tools/footprint_audit.py)	run: 183 PASS, 0 FAIL, 3 OPEN, 0 UNRECOGNISED, 25 fabricated features over all 211 designators
collision check (tools/collision_check.py)	run: 15 findings. Three of its four cases are open. See 1.5
.kicad_dru bracket check	run: 37 rules, balanced
STEP structural check	run: all files closed, no dangling entity reference. No STEP reader was available , so no CAD kernel has opened them
tools/simulate_production.py, after	run: 193 passed, 0 failed, 7 open

1.5 The collision check, in full

Case	Result
A – carrier parts against the plate underside	clear. Tallest is J15 at 10.00 mm against a plate underside at 18.0 mm
B – the DevKit against the plate opening	OPEN. The opening is 2 mm too short. J6/J7 are 1x22, so the pin span is 53.34 mm and the ESP32-S3-DevKitC-1 PCB is 63.0 mm: it needs carrier Y 3.2 to 66.2 mm and MP-01's opening runs 4.0 to 65.0 mm
C – modules against each other and jumper reach	OPEN. Six of twelve cannot be placed, and four cannot reach their connector within the 60 mm of ICD-EEG-006 section 3.2 from any position on the plate – M4 to J10 is 86 mm at best, M5 to J11 is 72 mm at best. That 60 mm is a crosstalk limit
D – plan area	OPEN. 17 356 mm² of modules into 12 409 mm² of plate: 140 % fill

Case D does not rest on the assumed envelopes. The two ADS1299 boards alone are 7 345 mm², **59 % of the net usable plate**, and that figure is the Raspberry Pi HAT

mechanical standard rather than a guess: the PiEEG-8 is a Pi shield. **Halving every other envelope still leaves 12 351 mm² against 12 409 mm².**

1.6 Reproducibility

Two consecutive runs of `tools/emit_all.py` produce **byte-identical SHA-256 checksums across all 51 files** of the handover set. Every uuid in the schematic is uuid5 over a fixed namespace, so a regeneration is diffable against the last one.

2. Where the repository contradicted the brief

#	The brief said	The repository says
1	the simulator's last run was 193 passed, 0 failed, 6 open	On this machine the untouched tree printed 186 / 0 / 5 . Six checks need cadquery and one open item needs trimesh; installing trimesh gave 187 / 0 / 6, and cadquery could not be installed. The register's 193 was obtained on a machine with cadquery. The run behind this work prints 193 / 0 / 7 and is <i>not the same</i> 193: six checks were added and six lost, and they cancel. A machine with cadquery would print 199
2	"four op-amp packages moving from SOIC to TSSOP"	Three. U1, U2 and U3. There is no U4; U7 is the SOT-23-5 comparator
3	the Murata part number should be corrected "if it does not encode 600 Ω "	The part number does encode 600 Ω – Murata's own numbering page reads the three figures as two significant digits and a count of zeros, so 601 is 600 Ω . The reviewer's reading was wrong. The defect is elsewhere and is real: PG is the large-current series and it is not made at 600 Ω in 1608, whose range stops at 470 Ω . BLM18PG601SN1D is not a part. Corrected to BLM18AG601SN1D
4	three BOM errors	Four. Omron B3F-4055 is a 12 x 12 mm switch on a 6 x 6 mm land pattern, and at 260 gf it also misses AVL-EEG-017's own 160 ± 50 gf window. Found by the audit the brief asked for, not by any reviewer. Corrected to B3F-1052

#	The brief said	The repository says
5	ICD-EEG-006 section 1 is the one home of the connector table	Four of its coordinates disagreed with design.py: J4 (41.0, 36.0 against 41.0, 50.0), J9 (66.0, 78.0 against 114.0, 62.0), J22 (30.0, 116.0 at 90° against 15.7, 73.5 at 270°) and J26 (128.0, 104.0 against 117.0, 8.0). Two moved at ECO-EEG-018 and the table never followed. design.py governs; the table is corrected and RUL-EEG-021 ruling 6 is added
6	design.py carries 68 kΩ (ECO-EEG-024)	The value did. design.py's NOTES, which are printed onto the fabrication and assembly drawings, still said 47 kΩ ; also said the analogue rails arrive at "J2 pins 11-14", where J2 has ten ways and they arrive at J23; and said the keep-outs are clear "on both layers" on a four-layer board. All three corrected
7	emit kicad/EEG-CAR-01_RevC_unrouted.kicad_pcb and EEG-CAR-01_RevC.kicad_pro	Those two names are incompatible. A KiCad project expects its board to share the project's stem; a differently named board opens standalone, without the net classes, the 156 assignments or the DRC severities. The board is EEG-CAR-01_RevC.kicad_pcb. See open question 1
8	the rule sheet is one page	It is three . See open question 2
9	five release-state paragraphs say "released for review"	Sixteen documents did , sixty-two occurrences. The five named are corrected, and so are the other six live statements in the four documents this round re-issued. Eight documents it did not re-issue still carry it. See open question 3
10	the DRC report is the authority for the routed result	It is, and it is authoritative about the rule set it had . RUL-EEG-021 ruling 5 now says that a DRC report states the rule set it measured against

3. Assumptions

Every one is registered or flagged. None is a sign-off.

#	Assumption	Where it is registered	Confidence
1	The twelve module envelopes, and their placement on MP-01	ASM-EEG-023 Rev B, MECH-D6-MODULE-ENVELOPES , new and blocks a build	judgement. No vendor drawing was consulted for any of them. Two are form-factor derived: the PiEEG-8 is a Pi HAT (65.0 x 56.5 mm) and the Pololu S13V15F5 is a 0.5 x 0.8 inch board
2	ELECTRODE minimum width 0.25 mm, above the board's 0.20 mm floor	tools/rules.py, flagged confirm; printed in LAY-EEG-034 section 5	judgement. Open question 4
3	ELECTRODE on L1 only, no vias	same	DSN-EEG-003 section 3.3 rule 3 read literally. Open question 4
4	ANALOGUE_REF minimum width 0.30 mm	same	impedance, not current. Open question 4
5	Pad entry within 25 % of the pad's minor dimension	tools/rules.py; LAY-EEG-034 section 6	judgement: finding 4 has no number in it
6	The schematic sheet split, nine sheets	tools/emit_kicad_sch.py, checked to be a partition of design.C	judgement
7	KiCad 8 – (kicad_sch 20231120), (kicad_pcb 20240108)	the handover README, with an offer to re-emit	assumed, as the brief instructed. Open question 5
8	Carrier component heights per footprint class	tools/mech_bodies.py, each with its source	datasheet, except the DIN 42802 socket at 10.00 mm, which has no qualified part
9	The DevKit stands 13.6 mm and its PCB is 63.0 x 25.5 mm	ICD-EEG-006 section 4 (<i>calculated</i>) and Espressif's published outline	derived

4. Open questions

1. The board file name. The brief asked for EEG-CAR-01_RevC_unrouted.kicad_pcb beside EEG-CAR-01_RevC.kicad_pro. A KiCad project expects board and schematic to

share its stem, and a board opened standalone loses the eight net classes, the 156 assignments and the severity that makes `track_dangling` an error – most of the specification. *Recommended default, and what is implemented:* the board is EEG-CAR-01_RevC.kicad_pcb, and that it is unrouted is said in its title block, its comments, LAY-EEG-034 and the README.

2. The rule sheet is three pages, not one. Compressing it means dropping the derivation of each net-class number, and a rule whose reason is invisible is a rule that gets relaxed at the first difficulty. *Recommended default:* leave it at three. If one page is needed for the order paperwork, sections 5 and 6 alone fit on one and the rest becomes an annex.

3. Eight documents still say the data is “released for review”. ASM-EEG-007, PARTS-EEG-019, QP-EEG-010, REG-EEG-012, RFQ-EEG-001, RISK-EEG-011, SVC-EEG-013 and TST-EEG-004. Three of them are out of scope by instruction (RISK-EEG-011, REG-EEG-012 and the research documents), and RFQ-EEG-001 additionally carries E-26’s self-contradicting switch class. *Recommended default:* one sweep ECO covering the five that are not out of scope, plus the RFQ at its next issue. It is mechanical and it is the defect RUL-EEG-021 section C ruling 4 forbids.

4. Four net-class numbers need your confirmation before the order goes out, because they constrain the contractor and they are this programme’s proposals, not transcriptions: the ELECTRODE minimum width of 0.25 mm; ELECTRODE restricted to L1 with **no vias at all**; the ANALOGUE_REF minimum width of 0.30 mm; and the 25 % pad-entry tolerance. *Recommended default:* confirm all four. The electrode restriction is the one worth a second look – it is DSN-EEG-003 section 3.3 rule 3 read literally and it is achievable on the present placement, but if the contractor says it forces a respin, relaxing it is a decision and not a concession to be made silently.

5. Which KiCad version is JLCPCB on? Everything is emitted for KiCad 8. Re-emitting for another major version costs nothing because it is generated. *Recommended default:* ask Lyle He before the order and re-emit if the answer is not 8.

6. MP-01 cannot carry twelve modules. 140 % fill, and the plate’s DevKit opening is 2 mm short. Four routes would each move it – a larger plate; a second tier of standoffs; an ADS1299 breakout that is not a Pi HAT, which AVL-EEG-017 section 2 M1 permits under qualification even though the *device* may not change; or moving the modules whose jumpers cannot reach onto the carrier as fitted parts. **None is decided and none is costed.** *Recommended default:* buy one of each module and measure it before deciding anything, because ten of the twelve envelopes are assumptions. The two that are not – the ADS1299 boards – are 59 % of the plate on their own, so **the measurement most likely to settle it is the PiEEG-8’s actual outline.**

7. Should the in-house router produce a reference route for comparison? It was not done: the brief puts it last and everything ahead of it took the time. *Recommended default:* no, until the contractor’s placement is in. A reference route against a placement nobody has confirmed compares two guesses.

8. `firmware/main/board_pins.h` is still hand-maintained against a check in `tools/simulate_production.py`. Untouched here, as instructed; no pin assignment moved, so it did not need to move. *Recommended default:* leave it until a change forces a pin move, then generate it.

5. The handover set

kicad/RevC_layout_inputs/ – **51 files, 2.6 MB**, with SHA256SUMS.txt carrying a SHA-256 and a one-line description for each. The manifest is the authority and is not reproduced here.

Group	Files
project	EEG-CAR-01_RevC.kicad_pro
schematic	EEG-CAR-01_RevC.kicad_sch + nine sheets, EEG-CAR-01.kicad_sym
board	EEG-CAR-01_RevC.kicad_pcb, EEG-CAR-01_RevC.kicad_dru
netlist and BOM	EEG-CAR-01_RevC-IPC-D-356A.ipc, EEG-CAR-01_RevC_BOM.csv, footprint_audit_RevC.md
placement	two provisional CPLs, EEG-CAR-01_RevC_outline_and_fixed_connectors.dxf
3D	22 footprint bodies under step/footprints/, three assembly bodies under step/
evidence	EEG-CAR-01_RevC_schematic_netlist_check.txt, EEG-CAR-01_RevC_collision_check.txt
rules	LAY-EEG-034_RevA_carrier_layout_rule_sheet.pdf and .md
covering note	README_layout_inputs.md, SHA256SUMS.txt

Nothing was sent. No tool in tools/ transmits anything.

6. Commits

Commit	Work package
c652258	WP1 revision bookkeeping and the Rev B withdrawal ECO (ECO-EEG-030)
34feea1	WP2 BOM and footprint corrections (ECO-EEG-031)
553bec1	WP3 the seven findings encoded as rules, and Rev B regraded (ECO-EEG-032)
e15724c	WP4 the native KiCad schematic (ECO-EEG-033 part 1)
b142243	WP5 unrouted board, project file and DXF (part 2)

Commit	Work package
9e96ce1	WP6 3D bodies and the collision check (part 3)
0fb4e9d	WP7 LAY-EEG-034, the layout rule sheet (part 4)
4b87f44	WP8 the handover set and its README (part 5)
385429a	WP9 document register close-out
17c02ad	WP9 (cont) release-state statements in the four re-issued documents

Documents advanced a letter: DSN-EEG-003 to **D**; ECO-EEG-016, ICD-EEG-006, JIG-EEG-009 and AVL-EEG-017 to **C**; FW-EEG-001 to **D**; ASM-EEG-023 and RUL-EEG-021 to **B**. **LAY-EEG-034 is new at Rev A**. 218 citations and 24 file names followed, by `tools/revision_bump.py`.

Nothing in this package has been manufactured or measured. No safety engineer has reviewed this design. No layout engineer has seen Rev C, because Rev C has no layout yet.